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12 - STD	QUARTERLY EXAMINATION - 2025	
TIME :3.00 Hrs	CHEMISTRY	MARKS : 70

I. Choose the correct answer: **15 x 1 = 15**

1. Which of the metal is extracted by Hall-Heroult process?
   a) Al b) Ni c) Cu d) Zn
2. Carbon atoms in fullerene with formula C_{60} have
   a) sp^3 hybridised b) sp hybridised
   c) sp^2 hybridised d) partially sp^2 and partially sp^3 hybridised
3. The General formula for phyllo silicate is
   a) $(Si_2O_5)_n^{2n-}$ b) $(Si_4O_{11})_n^{6n-}$ c) $(SiO_4)^{4-}$ d) None of these
4. XeF_6 on complete hydrolysis produces
   a) $XeOF_4$ b) XeO_2F_2 c) XeO_3 d) XeO_2
5. The molecular formula of Marshall acid is
   a) H_2SO_5 b) $H_2S_2O_6$ c) $H_2S_2O_8$ d) $H_2S_2O_7$
6. Which of the following metal ion has the magnetic moment value is 3.86 BM
   a) Cr^{3+} b) Mn^{4+} c) V^{2+} d) All of these
7. The radius of an atom is 300pm, if it crystallizes in a face centered cubic lattice, the length of the edge of the unit cell is
   a) 488.5pm b) 848.5pm c) 884.5pm d) 484.5pm
8. The addition of a catalyst during a chemical reaction alters which of the following quantities?
   a) Enthalpy b) Activation energy c) Entropy d) Internal energy
9. After 2 hours, a radioactive substance becomes $(1/16)^{th}$ of original amount. Then the half life (in min) is
   a) 60 minute b) 120 minute c) 30 minute d) 15 minute
10. The pH of an aqueous solution is Zero. The solution is
    a) slightly acidic b) strongly acidic c) neutral d) basic
11. Which one of the following is not Lewis base
    a) ROH b) RNH_2 c) CaO d) SF_4
12. Williamson synthesis of preparing dimethyl ether is a / an
    a) S_N1 reactions b) S_N2 reaction
    c) electrophilic addition d) electrophilic substitution
13. Which of the following oxidizing agent is used to convert glycerol to meso oxalic acid
    a) Con. HNO_3 b) $Bi(NO_3)_3$ c) $KMnO_4$ d) LTA
14. The reagent used to distinguish between acetaldehyde and benzaldehyde is
    a) Tollens reagent b) Fehling's solution
    c) 2,4 - dinitrophenyl hydrazine d) semicarbazide
15. Which one of the following reduces Tollens reagent
    a) formic acid b) acetic acid c) benzophenone d) none of these

II. Answer Any Six Questions. (Q. No 24 is Compulsory) **6 x 2 = 12**

16. What are the steps involved in extraction of pure metal from their ores ?
17. Define inert pair effect
18. How will you prepare bleaching powder?
19. Out of $Lu(OH)_3$ and $La(OH)_3$ which is more basic. why?
20. What is pseudo first order reaction? Give an example.
21. Define ionic product of water. Give its value.

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22. How ethene react with Bayer's reagent?
23. What is formalin ? Give its use.
24. Atoms x and y forms bcc crystal structure. Atom X is present at the corners of the cube and y is at the centre of the cube. What is the formula of the compound?

III. Answer Any Six Questions. (Q. No 33 is Compulsory) **6 x 3 = 18**

25. How Silver is refined by electrolytic process?
26. What is Catenation? Write their properties.
27. Write note on Holme's Signal.
28. State Hume Rothery rule.
29. what is metal excess defect ? Give an example.
30. Differentiate order and molecularity of a reaction
31. Write Swern Oxidation reaction
32. What is buffer Solution? Give its type with example.
33. $CH_3COCl + H_2 \xrightarrow{Pd/BaSO_4} A \xrightarrow{NaOH} B \xrightarrow{\Delta} C$.

    Identify A, B and C

IV. Answer All The Questions. **5 x 5 = 25**

34. a) Explain zone refining process

    (OR)

    b) i) How will you convert boric acid to boron nitride?
    ii) Write the uses of Silicones.
35. a) i) Show that H_2SO_4 is a dehydrating agent.
    ii) Find out the hybridisation for the following compounds a) BrF_5 b) BrF_3

    (OR)

    b) Differentiate Lanthanoids and actinoids.
36. a) i) Write Arrhenius equation and explain the terms.
    ii) Write any three examples for the first order reaction.

    (OR)

    b) Derive the calculation of Packing efficiency for bcc crystal.
37. a) state and explain Ostwald's dilution law.

    (OR)

    b) i) Write the limitations of Arrhenius concept.
    ii) How will you convert phenol to phenolphthalein ?
38. a) How will you distinguish primary, Secondary and Tertiary alcohol by Victor Meyer's method?

    (OR)

    b) i) What is Urotropine? How is it prepared?
    ii) Write tests for Carboxylic acid.

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